DESIGN BRIEF / OCTOBER 20, 2026

BATX
Battery-cell basket

A model-ready specification for a broad manufactured-cell price index.

Core instruction

Preserve four understandable market categories. Put detailed capacities, product models, and performance specifications underneath them as controlled observations, not as independently weighted headline markets.

1. Basket allocation

ID	Market category	Weight
LFP_STORAGE	LFP prismatic cells for stationary energy storage	40%
LFP_EV	LFP prismatic cells for electric vehicles	30%
TERNARY_PRISM_EV	Nickel-based ternary prismatic cells for electric vehicles	20%
TERNARY_POUCH_EV	Nickel-based ternary pouch cells for electric vehicles	10%
TOTAL	Manufactured lithium-ion cells	100%

The proposed mix is **70% LFP / 30% nickel-based ternary**, and **40% storage / 60% vehicle cells**. These are design allocations, not measured market shares or volatility-optimized weights.

Scope and exclusions

Measure manufactured cells, not their raw materials or the finished systems built from them. Exclude lithium carbonate, cathode powders, assembled packs, complete storage systems, used or second-life cells, and retail bundles. Cylindrical cells are excluded from version 1 until comparable industrial-price coverage is established.

Status and authority

The allocation and scope above are the working requirements for this handoff. The ticker BATX is provisional. Calculation mechanics below are a reference proposal; data access, historical coverage, internal series weights, and settlement rules are not yet launch-approved.

Design basis: the requester's revised four-category basket. Market-source verification is dated October 20, 2026; see page 4 and references on page 5.

Broad categories, matched comparisons

2. Separate the market definition from the sample

Use this hierarchy: **category > approved product or assessment series > eligible observation > normalized series > category index**. A 280 Ah storage cell and a 314 Ah storage cell can both inform the storage category without receiving permanent weights in the headline basket.

Broad categories do not authorize unrestricted substitution. Freeze the approved series and their internal weights in a versioned source manifest. One supplier appearing on more websites must not gain more influence.

Control	Required treatment
Product identity	Record chemistry family, application, casing, model or assessment code, capacity, voltage, performance class, and condition. Keep an observation in exactly one category.
Commercial basis	Record geography, currency, unit, tax basis, delivery terms, quantity tier, payment terms, and spot-versus-contract pricing basis. Compare like with like.
Provenance	Keep the original publisher or seller, the actual assessed market or quoted product, observation date, publication time, capture time, and a retrievable evidence reference.
Composition	Normalize each approved series against its own base. Do not average whichever listings are available on a given day. New or replacement series require versioned approval.

3. Collection basis

Proposed starting scope: China domestic industrial cell pricing in native CNY. Do not blend it with US retail prices or European distributor quotes. Preserve each source's commercial basis; a delivered price is not automatically equivalent to an ex-works price.

Use CNY per Wh when a valid conversion is possible. For per-cell quotes, use the documented rated energy of the exact cell. Do not guess nominal voltage or capacity. A consistent per-cell series may instead be normalized in its original units.

Normalization does not remove quality differences

Converting prices to CNY/Wh or indexing them to 100 does not make different performance classes, delivery terms, or product mixes interchangeable. Published category assessments may also change their internal sample; track methodology changes rather than assuming composition is fixed.

No return to the rejected design

Do not restore separate headline weights for 174 Ah, 158 Ah, 144 Ah, NMC811 21700 5 Ah, or similarly narrow specifications. Detailed eligibility belongs in the source manifest, not the four-category allocation.

Calculate movements, not mixed price levels

The following is a **proposed fixed-base arithmetic index**, supplied so a model can build an explicit, testable implementation. It is not a claim that the production settlement method has already been approved.

4. Reference calculation

Let P[c,s,t] be the eligible positive price for series s in category c at observation time t. Let t0 be the common base date, w[c] the category weight, and a[c,s] a precommitted within-category coefficient.

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R[c,s,t] = P[c,s,t] / P[c,s,t0]
C[c,t]   = sum_s( a[c,s] * R[c,s,t] )
BATX[t]  = 100 * sum_c( w[c] * C[c,t] )
sum_c(w[c]) = 1;  sum_s(a[c,s]) = 1 for each c
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Use w = (0.40, 0.30, 0.20, 0.10) in the order shown on page 1. Store category weights as **4,000 / 3,000 / 2,000 / 1,000 basis points**. Internal coefficients must be nonnegative; their values remain unspecified until the eligible panel is selected.

At the base date every R and C equals 1, so BATX equals 100. All required series need valid base observations under a common, documented timing rule. Missing inputs must not trigger silent reweighting.

These are **base-period coefficients**, not a promise that effective contribution shares stay constant after relative prices move. Do not silently substitute a daily rebalanced return index; that is a different methodology requiring approval.

5. Synthetic acceptance tests

Test, measured from the base date	Expected result
Every category unchanged	BATX = 100
Only storage rises 10%	BATX = 104
Every category rises 10%	BATX = 110
Storage +10%; LFP EV -5%; others unchanged	BATX = 102.5
A cheaper, unapproved model appears	No index change from its appearance
A required series disappears	Missing-data flag; no weight redistribution

Examples are hypothetical test vectors, not observed battery returns. Preserve decimal precision through calculation and round only at the documented output stage.

Currency and rebasing

The proposed index measures native-CNY price movements; settlement in USDC does not by itself add an FX return. Any currency-conversion exposure must be separately specified. Rebase or revise only through an approved new version, with an explicit continuity rule.

EVIDENCE / SOURCE COVERAGE

Use source coverage without overstating it

6. Verified starting points

Source	Verified coverage	What it does not establish
TrendForce [1]	All four proposed categories appear separately, quoted in RMB/Wh. The price page states a monthly update cadence.	Four series are not four independent publishers. The page does not resolve all product, quantity, delivery, or licensing details needed for settlement.
InfoLink [2]	A dated public report gives separate Chinese LFP storage-cell assessments for 100, 280, and 314 Ah.	Those narrower series are not automatically a broad storage benchmark or substitutes for EV prices.
Fastmarkets [3]	A published launch specification defines monthly 314 Ah LFP prismatic cells, ex-works domestic China, in yuan/Wh.	The methodology notice alone does not establish unrestricted access, a complete price history, or settlement-use permission.

Storage has more distinct publisher leads in this review. Independent coverage for each vehicle category remains unproven. Do not describe the four-category basket as launch-ready or multi-source until the exact series and permissions are verified.

7. Evidence and timing controls

Independence: attribute evidence to its originating publisher, seller, and ultimate data source where known. Syndicated copies count once. Multiple products from one publisher remain exposed to that publisher's methodology and availability.

Freshness: distinguish the assessment period, publication timestamp, capture timestamp, and oracle processing timestamp. Re-fetching a monthly price daily does not produce fresh daily evidence. Do not interpolate prices to manufacture daily volatility.

Missing or disputed observations: flag the condition and apply only a precommitted carry-forward, fallback, dispute, or settlement-delay rule. Until that rule is approved, do not manufacture a final settlement value or redistribute the missing series' weight.

Revisions and auditability: retain original captures and corrected vintages. A historical simulation must use only information available at the simulated time. Published corrections need a documented rule before live contracts exist.

Keep contract definitions immutable

For each open contract, freeze category weights, eligible sources and series, internal coefficients, base values, timing rules, and fallbacks. Replacement products or new sources belong to a future version unless the contract already contains an explicit transition rule.

Source existence is not proof of economically independent prices, stable methodology, sufficient volatility, or legal permission to redistribute data. Treat each as a separate verification task.

HANDOFF / MODEL TASK

Implement the architecture; expose the gaps

Instruction to the receiving model

Design a deterministic, auditable BATX prototype around the four categories and weights in this document. Preserve the broad category structure and controlled series-level observations. Do not invent observations, independent sources, historical returns, data rights, or approval of unresolved parameters. Separate verified facts, proposed defaults, and blockers.

8. Required outputs

A. Source manifest: map each category to exact series, original URLs or provider identifiers, product and commercial definitions, publication cadence, access conditions, data-use status, history coverage, and internal coefficients. Mark unknown fields explicitly.

B. Calculator and tests: implement the proposed fixed-base method or explain a clearly labeled alternative. Return category indices, total index, contribution diagnostics, evidence lineage, and missing/stale flags. Pass the synthetic tests on page 3.

C. Feasibility assessment: distinguish an analytical prototype from a settlement-ready system. Test actual common-window data before reporting one- or three-month returns, volatility, correlations, stale-price frequency, or sensitivity to panel composition.

9. Decisions still required before settlement

Approve the exact source panel and internal coefficients; base-date alignment; observation and settlement schedule; minimum evidence and concentration requirements; stale-data limits and fallbacks; source revision handling; numerical precision; and product-transition rules. Data access and use rights must be resolved for every required source.

Do not optimize weights for a dramatic historical chart. Preserve **40 / 30 / 20 / 10** unless the requester approves a new design. A wider product label is not a substitute for comparable, observable prices.

References and verification scope

These primary pages were inspected on October 20, 2026. They support the source map, not the proposed allocations, formulas, licensing status, or a claim of statistical validation. No historical dataset or live production implementation accompanies this brief.

[1] TrendForce. Li-Ion Battery Industry Chain Prices: Battery Cell & Pack. Monthly category coverage.

https://www.trendforce.com/price/battery-price/battery_cell_and_pack

[2] InfoLink. Lithium prices continue to fluctuate as average cell and system prices edge up. March 20, 2026.

https://www.infolink-group.com/energy-article/ess-spot-price-20260320

[3] Fastmarkets pricing team. Fastmarkets to launch China PV module, BESS battery cell prices. June 23, 2026.

https://www.fastmarkets.com/insights/fastmarkets-to-launch-china-pv-module-bess-battery-cell-prices/